YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 2 OF 10 · 40 MINUTES

Thought experiments 思想实验

Scientists can test an idea in their heads first (a thought experiment), then turn their curiosity into questions an experiment can answer.

- I can tell the difference between a testable and a not testable question.
- I can write a testable question and a prediction with a reason.

thought experiment 思想实验 · **prediction** 预测 · **reason** 理由 · **testable** 可检验的 · **evidence** 证据 · **fall** 下落 · **gravity** 重力 · **air resistance** 空气阻力

THE LESSON

Do Now	4	Shake, listen and weigh 2 sealed mystery cups; write observation and inference; Today slide.
Hook	3	Vote: heavy or light ball, which lands first? Predict with 'because'; start Galileo story.
Learn	7	Galileo storyboard; teacher book and paper drops, predict each; testable questions.
Do	18	Card sort: testable or not (8 min); worksheet: 3 testable questions with predictions (10 min).
Think and Reflect	8	Heavier things fall faster: always, sometimes, never? Pair talk, share, exit ticket, logbook.

Hands-on: Groups handle sealed mystery cups and sort question cards; only the teacher does the book and paper drops.

Collect or assess: Card sort checked in class (note groups with 3+ errors); 3 testable winter questions (Rubric strand 1); exit ticket.

SAFETY

- Choking risk: mystery cups stay taped shut and are never opened; nothing in mouths.
- Falling book: teacher only, from about 1 m onto table or floor; hands and feet clear.

WATCH OUT

- Heavier is not always faster: drop flat and crumpled sheets (same mass) and insist on 'sometimes'.
- Never reveal cup contents; keep the kit card (A to L key) on the teacher desk, not on tables.

Language: Band 1 sits by the language buddy. Traps: 'light' (not heavy), 'fall' (verb, not autumn), 'testable' (not an exam).

Next week: Week 3 spinners: wing length vs fall time. This week: print spinner card, mark 2.0 m and 1.8 m on walls.

PREP FOR ONE CLASS

before the lesson about 67 min (57 could be done by someone else); after about 26 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Order or gather cup contents, 24 opaque paper cups, masking tape, marker 10 min · technician
- [ ] Book printing: worksheets x24, A3 Galileo poster, kit card (A5) 5 min · reprographics
- [ ] Optional: add 6 A4 Galileo storyboard table copies to the print job optional reprographics

THE WEEK BEFORE

- [ ] Make 12 mystery cups: object in, second cup on top, tape, label A to L 15 min · technician
- [ ] Print sort cards on A4 white card: 6 sets x 2 sheets = 12 sheets 5 min · reprographics
- [ ] Cut out sort cards: 6 sets x 14 cards = 84 cards (guillotine) 15 min · technician
- [ ] Put each sort card set (12 questions + 2 headers) in an envelope 5 min · technician
- [ ] Find a hardback book; practise the paper-on-top drop once 5 min · teacher

THE DAY BEFORE

- [ ] Put up the A3 Galileo storyboard poster 2 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Set out 2 cups per table, role and frame cards, pencils; teacher desk kit 5 min · teacher

AFTER THE LESSON

- [ ] Clean-up: sort cards counted into envelopes, cups and logbooks to boxes 3 min · students
- [ ] Check cup tape; re-tape any loose cup before the next class 3 min · technician
- [ ] Read exit tickets: who needs the Do Now recap, who is ready for Stretch 10 min · teacher
- [ ] Record a language band (1, 2 or 3) per student in tracker column G 10 min · teacher

PRINT

- Worksheet x24 (4 pages, 2 sheets each, double-sided, 48 sheets)
- Galileo storyboard poster x1 (A3 landscape, single-sided)
- Question sort cards x6 sets (2 sheets A4 card each, 12 sheets)
- Galileo storyboard table copies x6 (A4, single-sided) (optional)
- Teacher copy: kit card with mystery cup key (A5)

GET OR MAKE (NEW THIS LESSON)

- Paper cups, opaque x24
- Cup contents: marbles x3, cotton ball, coin, paper clip
- Cup contents: eraser, dice, small bell, bottle cap
- Cup contents: small wooden block, dry rice (1 teaspoon)
- Masking tape x1 roll, marker pen x1
- Small envelopes x6 (for sort card sets)
- Hardback book, about A4 size x1
- Plain A4 paper x3 sheets (drop demo)
- Teacher storage box for the mystery cups x1
- Mystery cups x12, sealed, labelled A to L (make)

Have ready from earlier weeks: Class box, Scientist's Logbooks x24, Pencils x24, Group role card sets x6, Sentence frame card sets x6 (check), Classroom language cards x24, My Science Dictionary x24